The Monetary–Financial–Fiscal Macromodel (MFFM): A Global Projection and Scenario Tool

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July 2026 — first draft

Abstract

Policy questions cross borders, but the tools for answering them mostly do not: institutions either operate world models that require specialist teams, or stitch single-country analyses together by hand. This paper documents the MFFM, a platform built to close that gap. It produces internally consistent projections and scenarios for 55 economies in a web browser, fast enough to revise them live in a meeting, and everything it shows comes from one estimated model: a Bayesian vector autoregression for each economy, linked to its partners through three trade-weighted foreign variables and four global financial prices, estimated jointly across missing data, and solved as a single linear fixed point. One example carries the paper from a first click to a policy conclusion. Pinning a 100 basis point rise in the US corporate bond yield does not produce a spread shock, because in the data corporate yields rarely rise alone; the model raises government yields and the policy rate with it. Holding the ten-year yield fixed on impact isolates the credit shock; the global solve then carries it to every economy, deepening the world output loss to about 3.6 times what a US-only analysis would show, and a Smets–Wouters counterfactual prices the monetary easing that would undo the damage.

Keywords: scenario analysis, conditional forecasting, GVAR, Bayesian VAR, policy counterfactuals. **JEL:** C32, C53, E37, E47.

1 Introduction

When markets move, or a policy changes, or a shock hits somewhere in the world, policymakers need an answer within days: what does this mean for growth, inflation, and interest rates, at home and abroad? A useful answer is quantitative, carries its uncertainty with it, and is consistent — the view taken for the United States and the view taken for the euro area cannot rest on contradictory assumptions about the same world. Any central bank produces such answers for its own economy as a matter of routine. Producing them for many economies at once is the hard part, and the hard part is what this paper is about.

Scenario analysis is how policy institutions now confront this kind of question. Stress tests are built on scenarios (Adrian et al., 2020b); growth-at-risk made the conditional distribution of outcomes, rather than the point forecast, a standard object of surveillance (Adrian et al., 2019); and the research frontier treats scenarios themselves as statistical objects, to be checked for concordance with a reference forecast, weighted by plausibility, and synthesized into a single view of macroeconomic risk (Adrian et al., 2025). Behind all of this stands

*International Monetary Fund. The views expressed here are those of the authors and do not necessarily represent the views of the IMF, its Executive Board, or IMF management. We thank seminar participants at the MCM Financial-oversight-of-the-System meeting for comments.

one requirement. A scenario is a conditional statement about a joint distribution, and it is only as good as its conditioning: coherent across variables, so the narrative does not contradict the model, and coherent across countries, so one desk's downside is not another desk's baseline. Producing such scenarios — many of them, quickly, for many economies — is the unglamorous step on which the whole edifice rests.

That production step belongs today to specialists, organized in what the profession calls a suite of models. Central banks and international institutions maintain structural DSGE models for policy experiments, semi-structural workhorses for the projection process, and empirical models for forecasting, each run by its own team; the Eurosystem's own review documents the practice (Darracq Pariès et al., 2021), and the estimated open-economy models that central banks operate (Adolfson et al., 2007a), the IMF's Global Macrofinancial Model (Vitek, 2018), and the quantitative model of the Integrated Policy Framework (Adrian et al., 2020a) are all of this kind: powerful, and staffed. The nearest relative of our platform is the ECB's multi-country model, which extends the ECB-BASE semi-structural framework to the five largest euro-area countries and anchors the staff projections (Angelini et al., 2019, 2025); we return to it below, because its documentation sets the standard for papers of this genre. What none of these arrangements solves is reach. The distance between what a specialist team can supply and what every country desk needs is well documented from inside the profession (Lindé et al., 2016), so most desks improvise: single-economy models and judgment, reconciled across borders by iteration, with one desk's output becoming another's assumption and the aggregation living in spreadsheets. The result is slow, hard to reproduce, and only as consistent as the last round. Between the spreadsheet and the specialist model sits the gap this paper addresses: scenario production that any economist can do, for any economy, with cross-border spillovers that are consistent by construction.

The MFFM — the Monetary–Financial–Fiscal Macromodel — fills that gap by making every desk a producer. It runs in a web browser, with nothing to install and no code to write, and opens on a baseline forecast for each of 55 economies, estimated from the latest published data and covering growth, inflation, interest rates and financial prices, with a full fiscal block for most economies and uncertainty bands throughout. The 38 main blocs are linked to one another, the euro area entering as one bloc whose outcome its 17 members inherit. A scenario is built by *conditioning*: click a variable, drag its future path or import an outside forecast, and the model recomputes the most likely path of everything else, in every linked economy. Because the full cross-country solve takes about a second, a desk economist can build and revise scenarios in the meeting rather than commission them from a modeling team. In the language of the synthesis literature, the tool mass-produces the ingredients that scenario synthesis consumes: a reference forecast and coherent conditional scenarios, for every economy at once, ready to be weighed and combined.

One estimated model sits behind the screen. Each economy is a Bayesian vector autoregression; three trade-weighted foreign variables — partner demand, partner inflation, the partner short rate — and four global financial prices tie the economies together, a Bayesian global VAR in the tradition of Crespo Cuaresma et al. (2016). Sparse linkage of this kind is what makes a global model estimable at all, an insight the tool inherits from the global VAR literature. Two obstacles remain, and the design is built around them. The data are ragged — samples of different lengths, publication lags, a pandemic — so the model treats missing values as quantities to be estimated, not as reasons to discard data. And interaction demands speed, which the model delivers by algebra: once estimated, any scenario is a linear problem, so the world solution is computed once, shipped with the tool, and reused for every scenario a user tries.

One worked example carries the paper from a first click to a policy conclusion. Ask for a 100 basis point rise in the US corporate bond yield and the model surprises you: government yields and the policy rate rise too, because in the data corporate yields rarely rise alone. The surprise teaches the central craft of conditional forecasting — a scenario is defined as much by what you hold still as by what you move. One more pin, the ten-year Treasury yield held at baseline on impact, turns the same rise into a true credit shock: output falls, policy eases, debt accumulates. The global solve then carries the shock abroad, and a structural counterfactual — an unanticipated easing in the model of Smets and Wouters (2007), laid on top of the scenario — prices the policy response that would undo the output loss. The parts of the machine are borrowed, and Section 2 gives the credits while writing the model down next to the classical global VAR it descends from; the contribution is the integration — one estimable global model, current with the data, solved instantly, usable by any economist. Section 3 is the example. Section 5 concludes, and the appendices hold the algebra, the testing record, and the commands that reproduce every figure.

2 Methodology

Two models organize this section, and they are meant to be read against each other. The classical GVAR comes first, in the form the literature developed it, because the MFFM keeps its skeleton and the comparison is sharpest with both on the page. The MFFM follows: what a bloc is, how its foreign variables are built, how the system is estimated and then solved. The section closes with the question any design must answer — what this way of working buys, and what it costs — and with the US bloc, which runs through everything as the concrete example. Readers who want the tour first can jump to Section 3 and return.

2.1 The classical GVAR

The global VAR grew out of a practical need: to extend a national model with the foreign variables that national outcomes depend on, without attempting an intractable world system. Proposed by Pesaran et al. (2004) and developed fully in Dees et al. (2007), it models each economy as a VARX*, a VAR augmented with weakly exogenous foreign variables, and ties the country models into one global system through trade weights; Chudik and Pesaran (2016) survey the approach and di Mauro and Pesaran (2013) collect its applications.

Formally, country $i = 1, \ldots, N$ is described by a VARX*(p_i, q_i) model,

$$y_{it} \;=\; a_{i0} \;+\; \sum_{s=1}^{p_i} \Phi_{is}\, y_{i,t-s} \;+\; \sum_{s=0}^{q_i} \Lambda_{is}\, y^*_{i,t-s} \;+\; \sum_{s=0}^{q_i} \Upsilon_{is}\, d_{t-s} \;+\; u_{it}, \tag{1}$$

where y_{it} is the $k_i \times 1$ vector of domestic variables, y^*_{it} the country-specific foreign variables, d_t a small set of global variables (the oil price is the canonical member), and u_{it} the reduced-form innovation. The foreign variables are trade-weighted cross sections of the partners' variables,

$$y^*_{it} \;=\; \sum_{j=1}^{N} w_{ij}\, y_{jt}, \qquad w_{ii} = 0, \tag{2}$$

with fixed weights w_{ij}, usually bilateral trade shares.[1]

Two assumptions make this system estimable country by country. The first is *weak exogeneity* of y^*_{it} and d_t with respect to the parameters of bloc i:[2] each economy takes

[1] The weighting scheme matters less than one might expect; see di Mauro and Pesaran (2013, p. 169).

[2] Foreign variables may affect the domestic economy contemporaneously, but they are not affected by domestic disequilibria — the lagged domestic error-correction terms do not enter the marginal model of the foreign variables (Garratt et al., 2012, pp. 56–57). Careful applications test this restriction bloc by bloc rather than assume it.

the world as given. That is plausible for a small open economy and fails, by construction, for the largest one, so the United States is treated asymmetrically: it is the source of the global variables d_t, and its own foreign-variable set is restricted. The second assumption is a common estimation window. The blocs are estimated on a shared balanced sample, so ragged country coverage is resolved by truncating everyone to the quarters all blocs share.

In its full form the classical model is a *cointegrating* system. Each country model is cast as a VECMX*, the number of cointegrating relations is determined by Johansen-type rank tests, estimation is by reduced-rank regression conditional on the weakly exogenous variables, and over-identifying long-run restrictions can be imposed so that the estimated equilibria match theory (Garratt et al., 2012). The short-run coefficients are then estimated by least squares conditional on the long-run relations. This long-run structure is one of the GVAR's real strengths: it disciplines the medium run exactly where an unrestricted VAR is weakest.

The global solution comes from stacking. Collect $z_{it} = (y'_{it}, y^{*\prime}_{it})'$ and define the link matrices W_i, built from the weights, that extract bloc i's own and foreign variables from the global vector $y_t = (y'_{1t}, \ldots, y'_{Nt})'$, so that $z_{it} = W_i\, y_t$. Writing (1) at first order for exposition and substituting,

$$A_i W_i\, y_t \;=\; a_{i0} + B_i W_i\, y_{t-1} + u_{it}, \qquad A_i = (I_{k_i},\, -\Lambda_{i0}), \quad B_i = (\Phi_{i1},\, \Lambda_{i1}), \tag{3}$$

and stacking the N blocs,

$$G\, y_t \;=\; a_0 + H\, y_{t-1} + u_t \qquad \Longrightarrow \qquad y_t \;=\; G^{-1} a_0 + G^{-1} H\, y_{t-1} + G^{-1} u_t, \tag{4}$$

where G stacks the $A_i W_i$ and H the $B_i W_i$. The contemporaneous matrix G is invertible in applications, and inverting it *once* delivers the world reduced form, from which forecasts and generalized impulse responses (Pesaran and Shin, 1998) follow. Stability requires the eigenvalues of $G^{-1}H$ to lie in the unit disc, a property that is checked rather than guaranteed; applications routinely track how many eigenvalue moduli approach one, since near-unit roots signal the over-parametrization risks inherent in systems this large. Estimation in practice is supported by a mature toolchain (Smith and Galesi, 2014).

Figure 1 draws the pipeline: separate country estimations, stars from observed data, one stack, one inversion. That is the machine the MFFM descends from. Everything that follows keeps its two central ideas — sparse trade-weighted linkage and a stacked linear solution — and changes how the blocs are specified, estimated, and solved.

2.2 The MFFM country model

The MFFM strips the classical bloc down to a closed VAR and moves the difficulty elsewhere — out of the specification and into the estimation. Each of the $N = 38$ coupled blocs is a closed, constant-coefficient VAR(p),

$$y_{it} \;=\; c_i \;+\; \Gamma_i\, D_t \;+\; \sum_{\ell=1}^{p} B_{i\ell}\, y_{i,t-\ell} \;+\; u_{it}, \qquad u_{it} \sim \mathcal{N}(0, \Sigma_i), \qquad p = 3, \tag{5}$$

where $D_t = \big(\mathbf{1}\{q(t){=}2\},\, \mathbf{1}\{q(t){=}3\},\, \mathbf{1}\{q(t){=}4\}\big)'$ holds quarterly seasonal dummies, with $\Gamma_i \neq 0$ for the 18 blocs whose source series are published without seasonal adjustment and $\Gamma_i = 0$ elsewhere. The bloc vector carries the partition

$$y_{it} \;=\; \big(\, x'_{it},\;\; g'_t,\;\; x^{*\prime}_{it} \,\big)', \tag{6}$$

with x_{it} the domestic block (activity, prices, interest rates, credit, housing and, for 25 blocs, four fiscal drivers), g_t the four global financial prices (the WTI oil price, the US

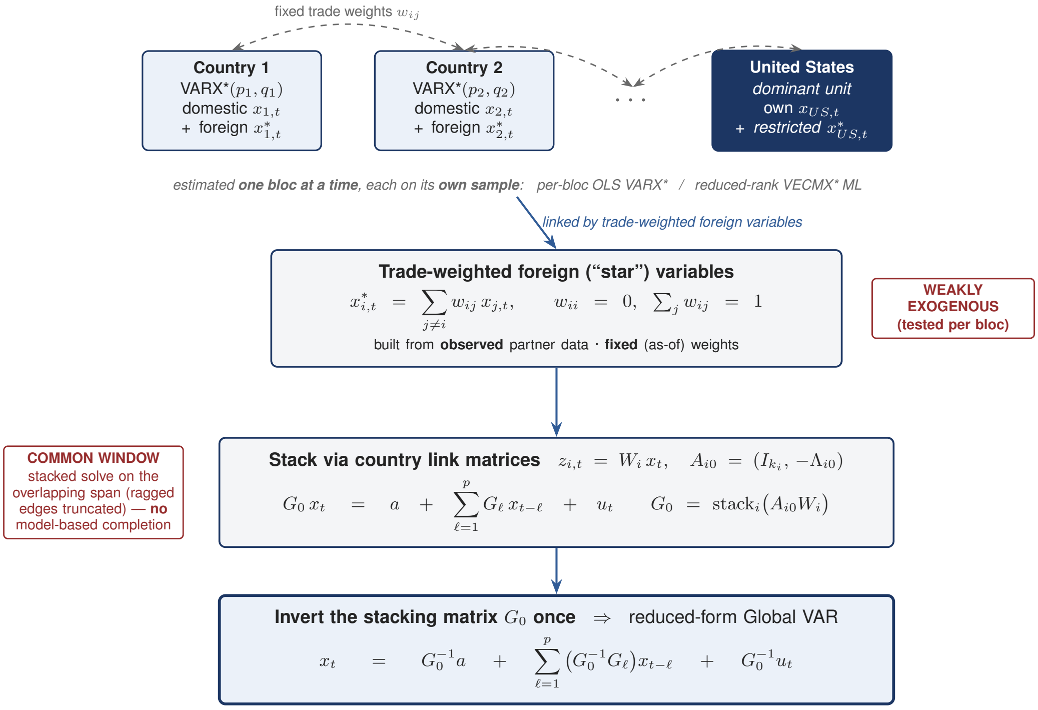

Figure 1: The classical GVAR pipeline. Country models are estimated one at a time on their own samples, linked by weakly exogenous trade-weighted stars built from observed data, then stacked through the link matrices and inverted once to give the world reduced form (4). The two red tags mark the assumptions the MFFM replaces: weak exogeneity and the common balanced estimation window.

ten-year Treasury yield, the US Baa corporate yield, and a US equity index), and $x^*_{it} = (\mathrm{ROWDEM}_{it}, \mathrm{ROWINF}_{it}, \mathrm{ROWFIN}_{it})'$ the three foreign variables. Variables tagged as logarithmic enter as $100 \ln$ levels, so the displayed quarter-on-quarter annualized growth is 4Δ; rates and GDP ratios enter untransformed. In the US bloc the four global prices are ordinary domestic columns — the United States is their source; every other bloc carries the same four series and receives their paths from the US solution.

The exogenous side is deliberately leaner than the Bayesian GVAR it is closest to. Crespo Cuaresma et al. (2016) give each country a near-full foreign mirror, $x^*_{it} = (y^*, \Delta p^*, eq^*, i^*_s, i^*_l, tc^*)'$ — trade-weighted counterparts of six of their seven domestic variables, all but the real exchange rate — together with the oil price as a global control determined inside the US model. The MFFM keeps only the three stars for partner demand, inflation and the short rate, and carries the financial linkage instead in the four *global* prices g_t: oil and three US-originated series (the ten-year yield, the Baa corporate yield and an equity index) that enter every bloc directly rather than as trade-weighted aggregates. That is the substantive difference between the two spillover mechanisms. Where Crespo Cuaresma et al. transmit foreign equity, long rates and credit through the weight matrix, the MFFM transmits US financial conditions as common prices, so a repricing at the source reaches every bloc's own equations at once — the mechanism behind the "US credit event as world credit event" reading of Section 3. The leaner star block trades the country-pair richness of a full foreign mirror for a sparser, more identifiable linkage and that common-price channel.

The defining difference from (1) lies in what is *absent*: there is no contemporaneous term $\Lambda_{i0} x^*_{it}$ and no weak-exogeneity restriction. The foreign variables are full endogenous columns with equations of their own, so each bloc is a closed VAR that can forecast its foreign environment unaided. Contemporaneous co-movement between foreign and domestic variables is carried by the covariance Σ_i rather than by a structural loading, and the cross-country consistency that weak exogeneity buys the classical model is imposed twice elsewhere: on the star *data* at estimation time (Section 2.3) and on the star *paths* at solve time (Section 2.4). One consequence deserves emphasis. Nothing in this design stops the United States from carrying foreign variables of its own, and it does: the classical asymmetry disappears, and the world feeds back into the US bloc as into any other.

The star data are built as in (2), with one refinement forced by the unbalanced panel. Let ℓ^g_{jt} and ℓ^π_{jt} denote partner j's $100 \ln$ real GDP and consumer-price levels and r_{jt} its short rate (the policy rate where one exists, the ten-year yield as fallback), all read from j's *completed* panel in the sense of Section 2.3. The three foreign variables of bloc i are

$$\mathrm{ROWDEM}_{it} = 4 \sum_{j \neq i} \tilde{w}^{\mathrm{DEM}}_{ij,t} \left(\ell^g_{j,t} - \ell^g_{j,t-1}\right), \tag{7}$$

$$\mathrm{ROWINF}_{it} = 4 \sum_{j \neq i} \tilde{w}^{\mathrm{INF}}_{ij,t} \left(\ell^\pi_{j,t} - \ell^\pi_{j,t-1}\right), \tag{8}$$

$$\mathrm{ROWFIN}_{it} = \sum_{j \neq i} \tilde{w}^{\mathrm{FIN}}_{ij,t} \, r_{j,t}, \tag{9}$$

the first two trade-weighted annualized growth rates, the third a level. The applied weights renormalize the fixed bilateral goods-trade shares w_{ij} (IMF merchandise-trade statistics, 2021–23 average) over the partners available and eligible at each date, per component:

$$\tilde{w}^k_{ij,t} = \frac{w_{ij} \, \mathbf{1}\{j \text{ eligible for } k \text{ at } t\}}{\sum_{j'} w_{ij'} \, \mathbf{1}\{j' \text{ eligible for } k \text{ at } t\}}, \qquad k \in \{\mathrm{DEM}, \mathrm{INF}, \mathrm{FIN}\}, \tag{10}$$

so a partner whose sample has not yet started lends its weight to the others instead of truncating everyone to a common window.[3]

One modeling choice remains, and it is the deepest difference between the two designs: the long run. The classical GVAR disciplines it with cointegrating restrictions; we discipline it with shrinkage. The MFFM works in levels under a Minnesota-type prior centered on univariate random walks, the Bayesian tradition for possibly integrated data that avoids pre-testing for ranks (Doan et al., 1984; Litterman, 1986; Sims and Zha, 1998). This shrinkage-in-levels route to the GVAR is itself established: Crespo Cuaresma et al. (2016) first replaced the classical cointegration step with Bayesian shrinkage on the country-model coefficients, and showed that hierarchical Minnesota-type and stochastic-search priors forecast an international macro-financial panel more accurately than a random walk with drift, an unshrunk global model, or isolated country VARs. The MFFM shares that estimation philosophy — exogenous trade-weight links, a VARX* per bloc, shrinkage in place of rank tests — and differs in what the estimated system is then put to: an interactive conditional solve (Section 2.4), a joint EM completion of the pandemic-holed panel, an endogenous anchor economy, and the fiscal and external accounting satellites, rather than density forecasting alone. The deliberate exceptions are the fiscal ratios — revenue, primary expenditure, the effective interest rate, the stock-flow term — whose prior is centered on white noise (own-lag coefficient zero): a GDP share is not a random walk, and a fiscal drift the data do not insist on should not be allowed to compound over a five-year projection. What is given up relative to the VECMX* is the explicit long-run structure; what is gained is that no rank and restriction decisions

[3] Eligibility differs by component: a bloc with no usable short-rate series drops out of FIN only, and Argentina's 2023–24 inflation episode, which would dominate the price and rate averages, excludes it from INF and FIN while it remains in DEM.

have to be made, and hold, for 38 blocs simultaneously. For blocs with $\Gamma_i \neq 0$ one further convention matters in daily use: the deterministic term is quarter-specific in history but averaged in projection — forecasts replace $c_i + \Gamma_i D_t$ by $\bar{c}_i = c_i + \frac{1}{4}(\gamma_{i,Q2} + \gamma_{i,Q3} + \gamma_{i,Q4})$ — so projected paths are seasonally adjusted while the model fits the unadjusted data it was given. A related accrual convention spreads coupon-lumpy government interest payments evenly within each calendar year for eight fiscal blocs; the online manual documents both.

2.3 Estimation

What makes estimation nonstandard here is not the prior but the data. The panel is ragged: bloc samples differ in length by more than a decade, and at any moment the latest quarter is published for some blocs and not others. The pandemic quarters 2020Q2–Q4 are treated as missing by choice, for the macroeconomic block only, so that three observations of unprecedented size do not dominate the estimated dynamics of thirty normal years (Lenza and Primiceri, 2022); the financial columns stay observed, and of the foreign variables ROWDEM and ROWINF are masked with the macro block while ROWFIN remains observed. And the regressors x^*_{it} are not data at all: they are functions of the partners' incomplete panels. Estimation therefore solves three problems jointly within each bloc — missing cells, hyperparameters, coefficients — and a fourth across blocs, because the stars each bloc needs are outputs of the other blocs' completions.

Within a bloc, everything runs through the state-space form. For given coefficients, stack $\alpha_t = (y'_{it}, \ldots, y'_{i,t-p+1})'$ and write

$$\alpha_t = \mathbf{T}_i\, \alpha_{t-1} + c_\alpha + \eta_t, \qquad y^{\text{obs}}_{it} = Z_{it}\, \alpha_t + \varepsilon_{it}, \qquad \varepsilon_{it} \sim \mathcal{N}(0,\, 10^{-12} I), \tag{11}$$

where $\mathbf{T}_i$ is the companion matrix of (5) and the selection Z_{it} keeps exactly the cells observed at date t, so missing cells simply drop out of the measurement equation. The simulation smoother of Durbin and Koopman (2002) then draws the missing cells exactly, conditional on everything observed — data augmentation in the sense of Tanner and Wong (1987).

The prior on (β_i, Σ_i), where β_i stacks $(c_i, \Gamma_i, B_{i1}, \ldots, B_{ip})$, is the natural-conjugate normal–inverse-Wishart family with the hierarchical Minnesota structure of Giannone et al. (2015). Conditional on Σ_i, the prior mean of each equation is the univariate random walk — own first-lag coefficient one — except on the fiscal-ratio columns, where it is white noise; the prior variance of the coefficient on lag ℓ of variable j is proportional to $\lambda_i^2/(\ell^2\psi_j)$, with the scales ψ_j fixed at univariate autoregressive residual-variance estimates. The sum-of-coefficients dummy observations of Doan et al. (1984) discipline long-run drift with tightness μ_i; on seasonal blocs the dummy-observation entries for the D_t columns are zeroed and the corresponding prior rows left diffuse, so the seasonal pattern is estimated essentially from the data. The two tightnesses are estimated, not calibrated: λ_i carries a Gamma hyperprior with mode 0.2 and standard deviation 0.4, μ_i mode 1 and standard deviation 1 — the settings of Giannone et al. (2015) — and both are updated by random-walk Metropolis–Hastings on the marginal likelihood of the completed data, with proposals reflected into $[10^{-4}, 5]^2$. The single-unit-root dummy is not used.

The full system is estimated by the following iteration.

1. *Initialize.* Build starting stars from the raw partner panels (leading gaps back-filled with first valid values); set (λ_i, μ_i) at their marginal-likelihood modes.

2. *Bloc pass (data augmentation).* For each bloc i, holding its stars fixed: alternate (a) drawing the missing cells with the simulation smoother given (β_i, Σ_i); (b) drawing

(β_i, Σ_i) from the normal–inverse-Wishart posterior given the completed data; (c) updating (λ_i, μ_i) by Metropolis–Hastings. Retain the posterior-mean completed panel.[4]

3. *Star rebuild (Jacobi).* Recompute every bloc's foreign variables (7)–(9) from the partners' just-completed panels.

4. *Iterate* steps 2–3 until the largest change in any bloc's star history falls below 0.03 (excluding the pandemic-imputation window), with a cap of 12 outer iterations.

Each bloc's sampler runs 1,000 draws with the first 500 discarded; the shipped model is the posterior mean $(\hat{\beta}_i, \hat{\Sigma}_i)$ together with the 500 retained draws, from which the tool simulates its uncertainty bands. The scheme is an EM-type algorithm in the tradition of Dempster et al. (1977) with a Jacobi outer loop across blocs. When it stops, every completed history, every filled gap, and every foreign variable is a fixed point of the whole system — the estimation-time counterpart of the classical model's weak exogeneity.

2.4 The solved system

At solve time the question changes from "what are the parameters" to "what does the world look like under my conditions." A scenario is a set of pinned future cells — values a user imposes on particular variables in particular quarters — and the answer is the conditional forecast: the smoother of (11) run over the future, treating pins as observations (Waggoner and Zha, 1999; Bańbura et al., 2015). The near-degenerate measurement variance makes a pin hard; giving a cell a finite variance instead makes it soft, which is how outside forecasts enter at a chosen level of trust. For a single bloc, that is the whole story.

Coupled, one more requirement appears: bloc i's foreign variables must equal the trade-weighted average of what its partners actually do, and the partners' outcomes depend on their own foreign variables in turn. Because the smoother's conditional mean is *affine* in the imposed values, this consistency requirement is a linear fixed point. Stack into x the deviations from baseline of every bloc's three star paths over the $H = 20$-quarter horizon, augmented by the four global-price paths that the US bloc originates, so that $\dim x = N \cdot 3 \cdot H + 4H = 2{,}360$. Let d collect the direct impact of the user's pins on those objects, and let M be the cross-response operator: entry by entry, how one bloc's star deviation moves when a partner's solved paths move, built from the partners' conditional-forecast operators and the weights (10). Consistency is

$$x \;=\; Mx + d \qquad \Longrightarrow \qquad x \;=\; (I - M)^{-1} d, \tag{12}$$

with a unique solution if and only if $\rho(M) < 1$. The shipped system has $\rho(M) = 0.97340$, comfortably inside the unit circle: the margin $1 - \rho(M)$ is 2.66 percentage points, and it would take a uniform 2.73 percent strengthening of the entire coupling to reach $\rho = 1$. Nearness to one still warrants care, because the worst-case geometric amplification $1/(1-\rho)$ is 38 and steepens sharply toward the boundary — a re-estimation lifting ρ to 0.995 would more than double it — so $\rho(M)$ is recomputed and checked at every vintage, and a non-contracting one would not be shipped; the amplification actually realized in Section 3 is far smaller, the pinned shock decaying and the world aggregation reweighting it. The inverse is computed once per estimation vintage and shipped with the tool, so at run time a scenario costs one matrix–vector product plus one conditional forecast per bloc — which is why the browser

[4] Within the outer loop the smoother's companion matrix is held fixed at the previous pass's posterior mean and common random numbers are used across passes, which makes the outer iteration contract in practice. An exact data-augmentation pass, with the companion rebuilt at every sweep, is available as an optional final polish.

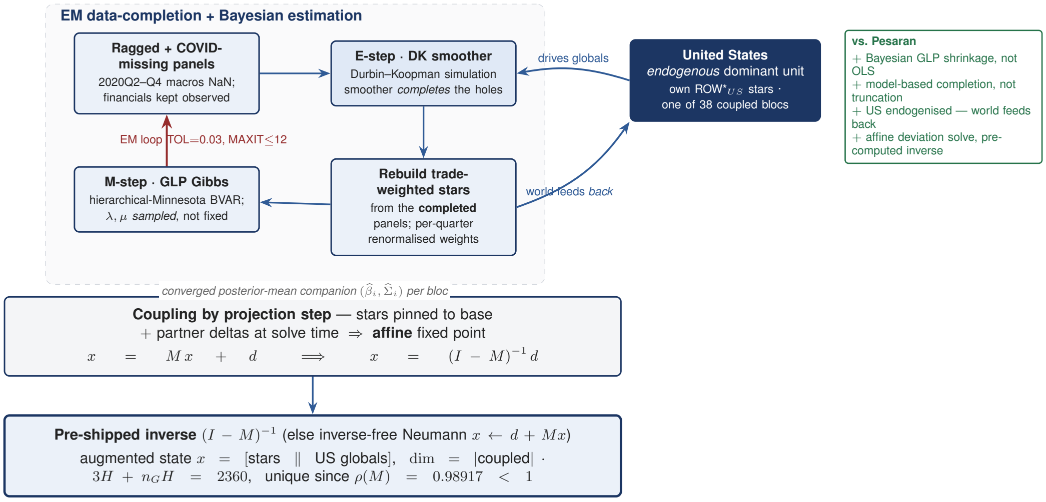

Figure 2: The MFFM pipeline. The left box is the estimation loop of Section 2.3: complete the ragged panels with the simulation smoother, re-estimate under the hierarchical prior, rebuild the stars from the completed panels, and iterate to a fixed point. The United States sits inside the loop as an ordinary coupled bloc: it originates the global prices and receives the world through its own stars. The converged companions then feed the affine solve of (12), whose inverse is computed once and shipped with the tool.

answers in about a second. Iterating (12) bloc by bloc (Gauss–Seidel) converges to the same point and serves as the verification path.[5]

The comparison with Section 2.1 now takes one sentence per object (Table 1, Figure 2). The classical GVAR stacks the blocs *contemporaneously* and inverts G once to obtain the world reduced form; the MFFM stacks the blocs *across the forecast* and inverts $I - M$ once to obtain the world scenario operator, which is the object a scenario tool needs. Weak exogeneity is replaced twice over, by the estimation-time fixed point in the star data and the solve-time pinning of the star paths. The common balanced window is replaced by model-based completion inside the estimation. Country-by-country reduced-rank regression under rank restrictions is replaced by hierarchical Bayesian shrinkage in levels. And the US asymmetry is removed: the United States carries foreign variables like every other bloc, and its four financial prices are broadcast to all blocs from its solved paths. The 17 euro-area member models sit outside this fixed point as satellites: each is a standalone BVAR whose euro-area columns (the common policy rate, area-wide demand and prices) are pinned to the EA bloc's solved paths, so members inherit the area outcome without enlarging the coupled system.

2.5 What the design buys, and what it costs

Why must the system be solved this way? Part of the answer is forced by the data, part is chosen for the tool, and part is a price paid knowingly. All three deserve to be on the table.

The data force the estimation design. At 38 blocs there is no common window worth estimating on: samples differ by more than a decade, so truncating every bloc to the shared span would discard most of the information in the long samples to accommodate the short ones — and the pandemic sits in the middle of whatever window remains. A classical VARX*

[5] Slowly, at this ρ: the residual shrinks like ρ^k, so hundreds of sweeps are needed — which is exactly why the inverse is precomputed rather than iterated live.

asks for more observations than a large bloc has in the overlap. Completing the panel inside the estimation, under shrinkage strong enough to make fifteen-year samples usable, is what makes a 38-bloc system estimable at all. The EM iteration is then a requirement rather than a refinement: the stars each bloc needs are functions of the other blocs' completed panels, so there is nothing to plug in — consistency between completion and estimation can only be reached by iterating to it.

The tool chooses the solution design. A scenario instrument must answer *conditional* questions in seconds, and the classical reduced form answers unconditional ones: G^{-1} delivers dynamics, not the response of a fifty-economy system to "hold this yield fixed while that one rises." The conditional-forecast operator is the object the tool needs, it is affine in what the user imposes, and affinity is what turns the coupled solve into $(I-M)^{-1}d$, computable once. The stars are pinned by deviation at solve time for the same reason: pinning preserves the linear structure on which interactivity stands.

The gains follow directly. Every economy is estimated on its full sample; the pandemic enters as missingness rather than distortion; posterior draws deliver bands, not just point paths; the United States sits inside the system, so the world feeds back; and between estimations the tool is updated by data operations — append a quarter, re-anchor the forecast — with the model untouched.

The costs deserve equal daylight. The long-run discipline of cointegrating restrictions is gone, replaced by priors and by user-set long-run anchors; the model will not discover a theory-consistent equilibrium on its own. Weak exogeneity is not tested but replaced by construction, so cross-country consistency is imposed rather than examined. The coupled multiplier stands on $\rho(M) = 0.97340$, close to one, which makes the global amplification the solve produces powerful and places weight on the star equations that generate it; Appendix B confronts the resulting responses with the impulse responses of a classical GVAR estimated on the same data. And the reduced-form layer identifies nothing by itself: without the structural overlay of Section 3.4, a scenario is a most-likely world, not a statement about cause and effect. The tool is built so that each of these costs is visible where it bites — anchors are set by the user, provenance is flagged on every chart, and the structural layer is drawn as its own line.

2.6 The US bloc, concretely

It helps to see all of this in one bloc, and the natural choice is the United States. The US model is (5) with $n = 40$ variables, $\Gamma_i = 0$ (US data are seasonally adjusted at source), and an estimation sample from 1998Q3 to 2025Q4 — 110 quarters. Its core follows the large US BVAR built at the Federal Reserve Bank of New York by Crump et al. (2025): real activity and its expenditure components, the labor market, prices, a term structure of interest rates, corporate bond yields, equity prices and their volatility, housing, and sentiment; their paper documents each series in detail. To this core the MFFM adds what its global role requires: the four fiscal drivers, the four global prices (which for the United States are simply domestic columns), and the three foreign variables. The full roster appears in Appendix C.

Partitioning $B_{i\ell}$ and Σ_i conformably with (6) makes the transmission channels explicit, and each does something different. The blocks $B_{i\ell}^{xx^*}$ put the three stars, at each of three lags, into every domestic equation, so a slowdown in partner demand works its way into next quarter's US output, investment, and employment the way any predictor would. The covariance block Σ_{xx^*} correlates foreign and domestic innovations, so a foreign surprise moves US variables *within* the quarter; when a scenario pins a star path, the conditioning machinery exploits exactly this. And the stars' own equations, the bottom rows of each $B_{i\ell}$, close the bloc, replacing the classical weak-exogeneity assumption and letting the same model run in two modes.

Table 1: The classical GVAR and the MFFM, side by side.

	Classical GVAR	**MFFM**
Country model	VARX*/VECMX* (1): contemporaneous foreign terms Λ_{i0}, cointegrating ranks, long-run restrictions.	Closed VAR (5) in levels: stars endogenous, co-movement through Σ_i, shrinkage instead of rank restrictions.
Estimation	Each country separately: reduced-rank regression conditional on weakly exogenous stars; least squares for the short run.	Hierarchical Bayesian per bloc (Giannone et al., 2015); (λ_i, μ_i) estimated by MH on the completed-data marginal likelihood.
Foreign variables	Built once from observed data; fixed weights; weak exogeneity assumed and tested.	Endogenous columns; rebuilt from partners' completed panels inside the estimation loop; per-quarter renormalized weights (10).
Missing data	Common balanced window; ragged edges truncated; pandemic handled with dummies.	Completed inside estimation (EM + simulation smoother); pandemic quarters treated as missing, financials kept observed.
United States	Source of d_t; restricted foreign-variable set; weak exogeneity fails for the dominant economy.	Fully endogenous: carries its own stars; the four global prices broadcast from its solved paths.
Global solution	Stack contemporaneously, invert G once (4): the world reduced form.	Stack across the forecast, invert $I - M$ once (12): the world scenario operator; $\rho(M) = 0.97340$.

Those two modes are the practical payoff of the design. Used alone, the way a desk economist uses a country model, the US bloc forecasts its own foreign variables; in effect it carries a reduced-form view of the world inside itself. In a global solve, its foreign variables are instead pinned through (12) to what the partner blocs actually produce, the partners' foreign variables see the United States in turn, and the fixed point reconciles everyone at once. The difference between the two modes is precisely the cross-border feedback, and it is measurable: Section 3.3 runs the credit-spread scenario both ways.

A last word on the fiscal side, because it works differently from the rest of the bloc and users notice. The VAR forecasts four fiscal *drivers* — revenue and primary expenditure as shares of GDP, the effective interest rate on the debt stock, and the stock-flow residual — while the debt-to-GDP ratio is not estimated at all. It is computed quarter by quarter from the government budget identity,

$$d_t \;=\; d_{t-1}\,\frac{1 + i_t/400}{1 + g_t/100} \;-\; \frac{pb_t}{4} \;+\; \frac{s_t}{4}, \tag{13}$$

where d_t is debt over annualized GDP, i_t the effective interest rate (percent per annum), g_t nominal GDP growth (percent, quarter-on-quarter, *not* annualized), pb_t the primary balance in percent of GDP, and s_t the stock-flow adjustment. Every scenario therefore carries its fiscal arithmetic with it: in a recession scenario revenue falls, growth denominators shrink, and the debt path steepens because accounting says it must, not because a debt equation was fitted. Section 3 shows this at work, and the same identity run backwards lets a user impose a debt target and read off the primary balance that would deliver it.

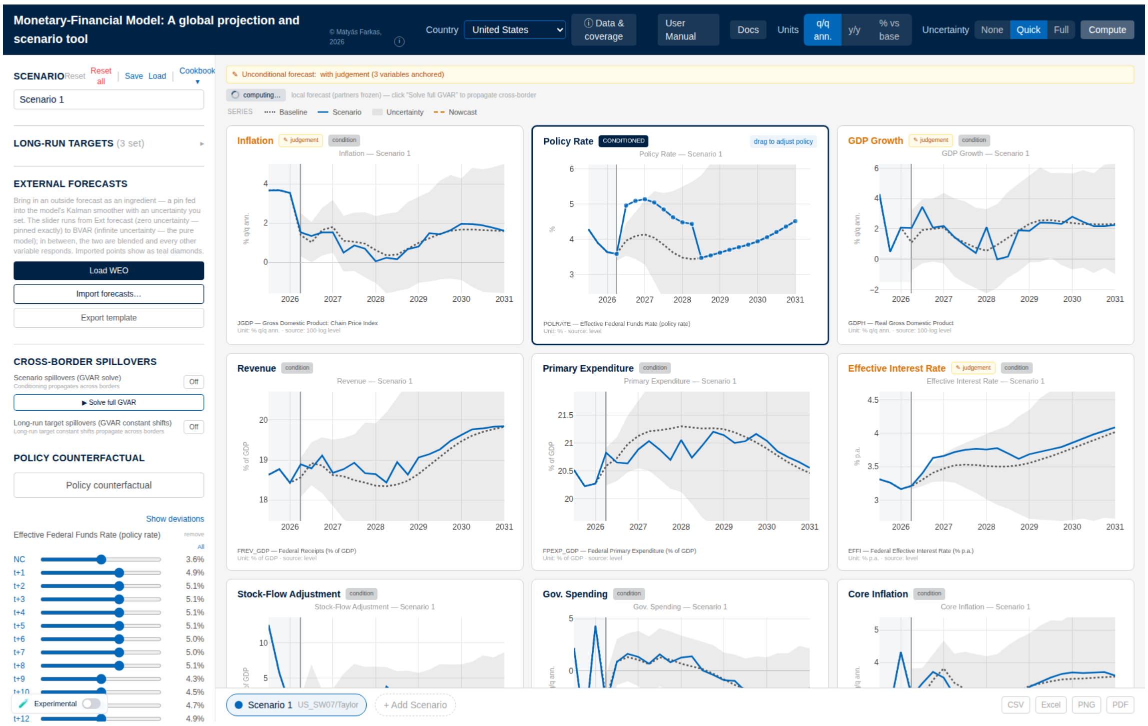

Figure 3: The tool during conditioning, US bloc. Each panel shows history, the baseline (dotted), the scenario (solid), and the uncertainty band; the highlighted panel carries the user's pins, and every other panel updates as they are placed. The left rail holds the scenario controls: long-run targets, external forecasts, the cross-border solve, and the policy counterfactual. Interface view from an earlier release; the layout is unchanged.

3 A worked example: a credit-spread shock and its global response

Credit has repriced. What happens, and what should policy do? We walk that one question through the tool, from the first click to a policy conclusion, and stop at each step to say what we did, what came back, and what it teaches. The response figures are generated by the tool's own computation engine, driven by scripts shipped in the repository (Appendix C gives the commands); the interface views show what the user sees while doing the same by hand.

3.1 One pin is not a spread shock

The experiment begins with the most natural move a user can make. We want the scenario in which the US corporate bond yield rises by 100 basis points, and in the tool that is a single action: open the Baa-yield panel, drag the nowcast quarter up by 100 basis points, and read the answer off the charts. Figure 3 shows the screen on which this happens.

The dashed lines in Figure 4 show what comes back, and it is not a credit event. The ten-year Treasury yield is up 69 basis points on impact, the policy rate climbs 45 basis points at its peak, and output barely reacts. The user asked for a spread shock and received a shock to the general level of interest rates.

Understanding why requires the central idea of conditional forecasting. The smoother does not treat a pin as a cause but as an observation, and it answers a precise question: among all the futures the estimated model considers possible, which is the most likely one in

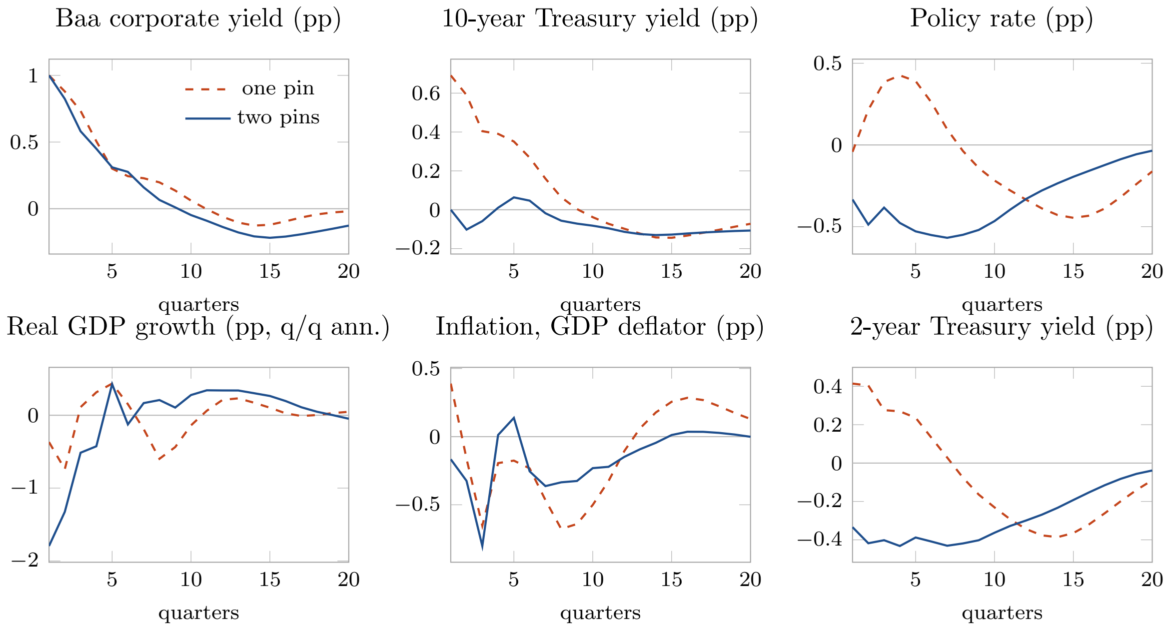

Figure 4: One pin is not a spread shock. Deviations from the baseline forecast after conditioning the US bloc on a 100 basis point rise in the Baa corporate yield at the nowcast quarter. With one pin (dashed), the model reads the rise as a broad tightening: the ten-year and two-year Treasury yields and the policy rate rise with it, and output is largely unaffected. Adding a single further condition, the ten-year yield unchanged on impact (solid), turns the same rise into a credit-spread shock: the curve stays put, output contracts, and policy eases. Horizontal axis: quarters from the nowcast.

which the Baa yield sits 100 basis points above baseline next quarter (Waggoner and Zha, 1999)? In 110 quarters of data, the corporate yield is rarely that much higher on its own. It is higher when the whole yield curve is higher. So the most likely world consistent with the pin is one of generally tighter financial conditions, and that is the world the model delivers: government yields up, the policy rate up, the spread between corporate and government yields barely changed. The model did exactly what it was asked. The user asked the wrong question.

3.2 Two pins make a risk shock

The fix costs one more click. Pin the ten-year government yield to stay at its baseline value in the same quarter (not lower, not higher, simply unchanged) and solve again. The only futures the smoother may now consider are those in which the corporate yield rises while the risk-free curve stands still, and in the data those are the episodes in which credit risk is being repriced.

The solid lines in Figure 4 show the difference. The spread now opens by the full 100 basis points. Output falls, with growth dipping 1.79 percentage points below baseline at its worst and the level of GDP bottoming out 1.02 percent below baseline in quarter 4. Inflation drifts down, and the policy rate eases, because in the estimated data that is what the Federal Reserve does when credit tightens and activity weakens. None of this was imposed. Two numbers were pinned; the other thirty-eight variables of the US bloc, every quarter of them, came from the model.

The episode carries two lessons. A scenario is defined as much by what you hold still as by what you move, so the craft lies in fitting the pins to the story, and stories need fewer

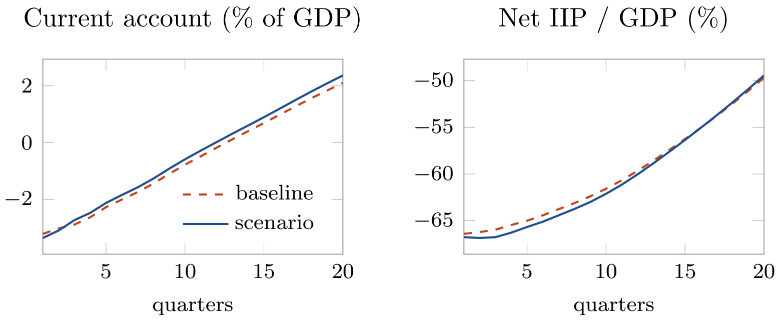

Figure 5: External-balance satellite: the credit-spread scenario's consequence for the US external accounts. Baseline (dashed) and two-pin scenario (solid) paths for the current account and the net international investment position, in percent of GDP. The current account is *anchored* to its real published value at the nowcast and moves only by the model's real net-export deviation; the net IIP is seeded from its real observed value and rolled forward by the balance-of-payments identity, neither estimated. The scenario improves the current account as the recession compresses imports, and the net-IIP ratio ends above baseline as the stronger balance rebuilds the stock. Horizontal axis: quarters from the nowcast.

pins than users expect. The second lesson is that the fiscal consequences arrive without any extra work: the recession lowers revenue and raises the debt ratio through the budget identity (13), leaving debt 1.9 percentage points of GDP above baseline after five years. The debt line moved because accounting moved it, and it sits on the same screen as everything else.

The external accounts follow the same way, from the balance-of-payments identity in place of the budget one. Each bloc is pinned by two real, sourced numbers: the net international investment position at the start—for the United States, -66.8 percent of GDP—and the current account itself, taken from its published value. The bloc's estimated exports and imports then supply only the *deviations* from that anchor, and the net position rolls forward by the balance-of-payments identity. Anchoring the level is what keeps the accounts honest: a current account built from real-volume net exports alone comes out wrong, and for several economies wrong in sign, once relative prices sit away from the chain-link reference. The tool therefore never reports that raw level, and it suppresses the few very-open blocs whose real-trade forecast would drive an implausible path. At the nowcast the current account is its published value—-3.73 percent of GDP for the United States, a surplus of 2.8 percent for the euro area and 1.89 percent for China.

Figure 5 traces what the credit-spread scenario does to them: the recession compresses imports faster than exports, so the current account improves by up to 0.27 percentage points and the net-IIP ratio ends 0.25 points above baseline at the five-year horizon as the stronger balance rebuilds the stock. The deviations do come from *real* trade, so a large terms-of-trade move would bias them—a channel to sharpen, not to distrust, now that the level is anchored to the sourced figure.

3.3 The global response

So far the rest of the world has been standing still. That was the point of working in a single bloc — the US model carries its own reduced-form view of its partners — and it is exactly what the global solve corrects. One button re-solves the same two-pin scenario as a fixed point of all 38 blocs at once, through the pre-computed operator of Section 2.4, and the answer arrives in about a second. Every economy in the dropdown now shows its own

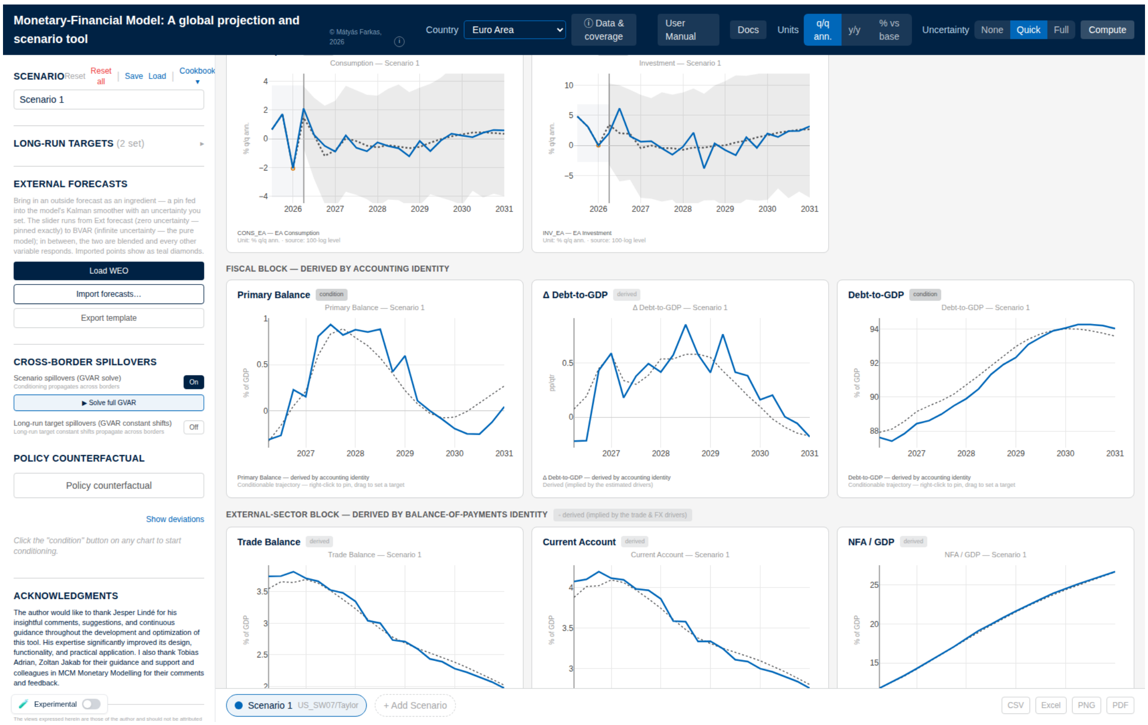

Figure 6: The euro-area screen with cross-border spillovers switched on. The scenario was built in the US bloc; the euro-area paths shown here come from the coupled solve. The lower rows are not estimated: the fiscal block follows from the budget identity (13) and the external block from the balance-of-payments identity, so a scenario's debt and current-account consequences appear without further input. Interface view from an earlier release; the layout is unchanged.

response, with the fiscal and external blocks derived alongside (Figure 6 shows the euro-area screen).

Figure 7 shows what changes, and the first thing to understand is *how* the shock travels, because it moves through two channels of different weight. The quieter channel is trade: weaker US demand reaches the partners through their foreign variables, and their weakness feeds back into the US stars, deepening the US growth trough from 1.79 percentage points on its own to 2.34 in the coupled solve. The louder channel is financial. The two variables we pinned are the model's global financial prices, carried by every bloc, so in the coupled solve the euro area and Japan face the same credit repricing the United States does, directly in their own equations. A US credit event is a world credit event in this model, by design rather than by accident, and the responses show it: euro-area growth troughs 2.15 percentage points below baseline, comparable to the US's own, and Japan deeper still at 3.48.

The world panel puts the two solves side by side. Treated as a US-only event, world growth dips 0.54 percentage points at its trough, which is simply the US decline scaled by its share of world output. The coupled solve deepens the world trough to 1.95 percentage points, about 3.6 times as much. That gap is what a country-by-country analysis leaves out, and a tool without a coupled global solve cannot draw the second line at all.

One qualification belongs with these magnitudes. Most of the gap between the two lines is definitional rather than estimated: the two pinned variables are global prices that every bloc carries in its own equations, so imposing them on the euro area and Japan is close to imposing the shock itself, and troughs of the same order as the US's own follow largely by construction. The genuinely estimated cross-border linkage is the quieter trade channel,

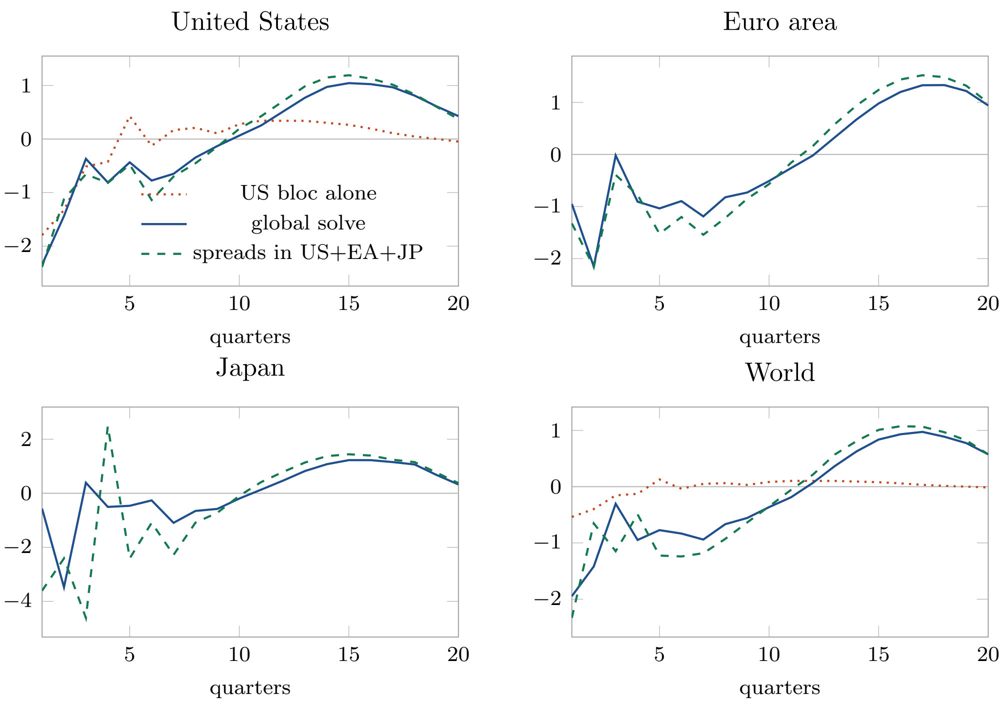

Figure 7: The same shock, three ways. Real GDP growth deviations (pp, q/q annualized) under the US credit-spread scenario of Figure 4. The dotted line solves the US bloc on its own, so the rest of the world stays at baseline by construction and world output moves only through the US share. The solid line runs the full coupled solve: the shock travels to the euro area and Japan through trade and financial prices, and their responses feed back. The dashed line imposes the same credit repricing in the United States, the euro area, and Japan at once. World output is the GDP-weighted aggregate over the coupled blocs, using the tool's world weights.

which on its own deepens the US trough only from 1.79 to 2.34 percentage points. The coupled solve's contribution here is internal consistency and speed across 38 blocs, not a measurement of international transmission — separating the common-price shock from its propagation would require an identification this descriptive exercise does not attempt.

The third line answers a question that comes up in every stress-testing exercise: what if the repricing is not only an American event? Imposing the same two-pin credit shock simultaneously in the United States, the euro area, and Japan deepens the world trough further, to 2.33 percentage points. Each economy now imports weakness from the others while generating its own, which is the configuration financial-stability exercises worry about. It takes six pins to build.

3.4 What should policy do?

Everything so far answers one kind of question: what is likely to happen. A policymaker's next question is of a different kind: what should we do about it? Conditioning cannot answer it. The smoother re-weights futures by likelihood; it does not identify cause and effect, and "the world in which the central bank cuts" is not the same object as "the effect of a cut" (Antolín-Díaz et al., 2021). For the second object the tool carries a structural layer: the policy-counterfactual option takes impulse responses from a fully specified DSGE model (here Smets and Wouters, 2007, from the Macroeconomic Model Data Base of Wieland et al., 2012) and lays the model-implied effect of a deliberate policy shock on top of the scenario.

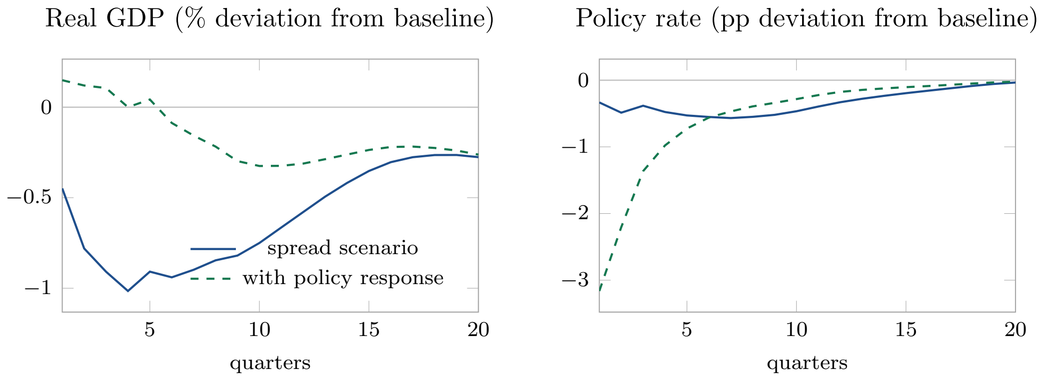

Figure 8: What should policy do? The credit-spread scenario (solid) and the same scenario with an unanticipated monetary-policy shock from the Smets–Wouters model layered on top (dashed), sized so that the easing offsets the output trough. Left: the level of real GDP relative to baseline; the counterfactual closes the gap the shock opened. Right: the policy-rate paths. The scenario's own rate falls because the model expects policy to respond; the counterfactual adds the deliberate surprise easing on impact, which then decays along the model's policy rule.

We ask a sharp version of the policy question: how large an unanticipated easing at the nowcast quarter would undo the scenario's output loss at its trough? The tool back-solves the required shock — in the interface, through the controls of Figure 9 — and the answer is 283 basis points. Figure 8 shows the result: the easing closes the GDP gap at the trough (left panel), and the policy rate jumps down on impact and then decays back along the Smets–Wouters policy rule (right panel), sitting below the scenario's own rate path throughout.

The size of that number is itself the finding. The scenario's rate path already eases, since the estimated model expects policy to respond to a credit tightening; the counterfactual asks for the *additional*, deliberate surprise that would fully offset the output trough, delivered in a single quarter. A one-off surprise is an inefficient instrument against a shock whose damage peaks a year out: its effect on output decays from the moment it lands, so offsetting a one-percent output loss takes several hundred basis points up front. A policymaker reading Figure 8 learns something concrete: against this scenario, a realistic response is either persistent rather than one-off, or it accepts part of the loss. Both variants are one click away in the tool, which is the point of having the layer interactive.

The last thing this experiment teaches is a boundary, and we want to state it plainly. The scenario layer is reduced-form: it says what tends to happen, given history, when spreads widen. The policy layer is structural: it says what a specific model implies a deliberate policy action does. The tool keeps the two visibly separate (the counterfactual is drawn as its own line, labeled with the DSGE model that produced it) because conflating them is the oldest mistake in applied macro. Appendix B returns to this distinction in more detail.

4 Outside forecasts, long-run anchors, and policy overlays

The worked example used only part of the platform, and the parts it did not use are the ones that connect the tool most directly to how forecasting is actually practiced in policy institutions. Institutional forecasts are never purely model output: they blend the model with outside forecasts, with views about where the economy settles in the long run, and with structural reasoning about deliberate policy choices. The MFFM carries a device for

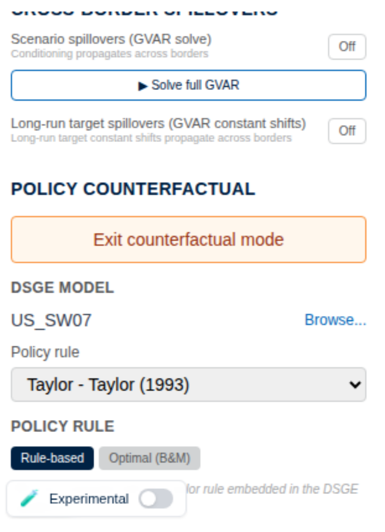

Figure 9: The policy-counterfactual controls. The user picks the DSGE model (here Smets–Wouters, US_SW07) and the policy rule from the Macroeconomic Model Data Base, enters counterfactual mode, and sets the policy shock; the counterfactual is drawn as its own line on every chart, labeled with the model that produced it. Interface view from an earlier release.

each of these, and because each one works through the same estimated model and the same smoother, the blended forecast remains a single coherent object rather than a collage. This section explains how the three devices work, in enough detail that a reader could predict what each one will do before touching it.

4.1 Outside forecasts as soft conditions

A desk economist rarely faces a blank slate. There is a WEO projection, a consensus number, or the desk's own view, and the practical question is never whether to use it but how much weight it should carry against the model. The tool answers that question with the one parameter the smoother already understands: measurement error. Recall from Section 2.4 that a scenario pin is treated as a future observation of the model,

$$y^{\text{pin}}_{j,t+h} = \left(Z\alpha_{t+h}\right)_j + \varepsilon_{j,t+h}, \qquad \varepsilon_{j,t+h} \sim \mathcal{N}(0,\, R_{j,h}), \tag{14}$$

and that a hard pin sets the variance $R_{j,h}$ essentially to zero, so the smoother matches the imposed value exactly. An outside forecast enters through exactly the same equation with a finite variance, chosen by a dial that runs from "impose exactly" to "ignore entirely." In between, the smoother forms the precision-weighted compromise between the outside number and the model's own predictive distribution, quarter by quarter, and every other variable in the system adjusts in proportion to how far the compromise moved. The device is the stochastic conditioning of Waggoner and Zha (1999) put behind a single slider, and its interpretation is exactly Bayesian: the outside forecast is data about the future, observed with an error whose size expresses how much the user trusts its source. Feed in a WEO path for US growth at high trust and the whole US bloc — and, through the solve of Section 2.4, the world — re-forms around it; lower the trust and the forecast relaxes back toward the model, continuously. Because imported paths are marked on the charts, a reader of the resulting scenario can always see which parts of it came from outside and how strongly they were held.

4.2 Long-run anchors

Users often know more about the destination than about the journey: the inflation target, an estimate of potential growth, a view on the neutral rate. The natural econometric home for such knowledge is Villani's steady-state parametrization of the VAR (Villani, 2009), in which the modeler places an informative prior directly on the long-run mean and the estimated dynamics carry the forecast toward it. The MFFM implements the reactive counterpart of that idea: the user states the long-run value — inflation at two percent, say — and the forecast is re-centered on it after estimation, so the device responds in seconds when the user changes her mind about the destination, with no re-estimation and no new draws. The mechanics respect the model's own dynamics. The stated long-run value fixes the terminal displacement of the anchored variable, and that displacement is phased in along the model's convergence profile,

$$\Delta y_{j,t+h} \;=\; w_j(h)\,\Delta_j^*, \qquad w_j(h) = \frac{S_h[j,j]}{S_H[j,j]}, \qquad S_h = \textstyle\sum_{k=0}^{h-1} \mathbf{T}^k, \tag{15}$$

where $\mathbf{T}$ is the companion matrix of Section 2.3 and Δ_j^* the terminal shift that lands the variable on its stated value. The weight $w_j(h)$ starts at essentially zero and accumulates to one at exactly the speed with which the model itself propagates a sustained change, so the near-term forecast is left untouched, there is no jump at the forecast origin, and the path glides to its new steady state neither faster nor slower than the estimated dynamics allow. The exact infinite-horizon steady state is not available here — under a prior centered on random walks the long-run matrix is near-singular — which is why the finite-horizon profile S_h/S_H stands in for the classical $(I - \mathbf{T})^{-1}$ and makes the device well defined for every bloc. A legacy alternative, a parallel shift that imposes the full adjustment from the first quarter as a constant drift, survives as a settings toggle, but the steady-state transition is the default and the shape users see. Uncertainty bands are unaffected, since the device moves the mean and nothing else; in the seasonal blocs the re-centering is applied identically across the four quarterly intercepts, so the anchor targets the seasonally adjusted path and leaves the seasonal pattern alone. Anchors can be set per variable or jointly, and the shipped defaults — official targets and consensus long-run values — are the ones the evaluation of Appendix B examines as a forecasting device in their own right.

4.3 The policy overlay: a DSGE delta on a reduced-form world

The counterfactual of Section 3.4 rests on a division of labor that deserves to be stated precisely, because it is the platform's answer to the oldest objection to reduced-form policy analysis. The estimated model supplies the world: the scenario, with everything it implies, is a statement about likelihoods under the policy behavior embedded in the data. What the estimated model cannot supply is the effect of a *deliberate departure* from that behavior — that is a causal object, and for it the tool turns to a structural model. The user selects a DSGE model and a policy rule from the Macroeconomic Model Data Base (Wieland et al., 2012); the platform ships the model's impulse responses to an unanticipated policy shock. When the user specifies a rate adjustment — a path, or a single surprise at one quarter — the layer back-solves the sequence of policy shocks that delivers exactly that adjustment under the chosen model's rate response, by inverting the convolution

$$\Delta r_{t+h} \;=\; \textstyle\sum_{k\le h} \phi_{h-k}^r\,\eta_k, \tag{16}$$

where ϕ^r is the rate's impulse response and η the shock sequence; the same shocks, run through the model's responses for inflation and output, give the deltas for those variables.

The deltas are then laid on top of the scenario paths — inflation accumulated into the price level, the output gap applied as a level deviation, conventions that matter and are easy to get wrong — and the smoother is run once more with these paths imposed, so that the remaining thirty-odd variables of the bloc adjust coherently around the structural core. The counterfactual appears as its own line, labeled with the model that produced it. The logic is a delta method: only the marginal effect of the deliberate policy surprise is asked of the structural model, which is precisely the piece the reduced-form scenario cannot identify, while everything the data speak to — the baseline, the scenario, the co-movements — stays with the estimated model. Different DSGE models give different deltas, and the tool makes switching between them a one-click sensitivity check rather than a research project.

5 Conclusion

What does the tool give an economist today? A morning's question — credit has repriced, what does it mean, what should we do — became, in Section 3, two pins, one button, and one structural overlay: a globally consistent scenario for 55 economies, its fiscal arithmetic included, and a priced policy response, all inside an hour that used to be a memo requesting a model run. The scenario-analysis literature of Section 1 presumes that coherent scenarios and reference forecasts exist and can be produced on demand; the MFFM is the machine that produces them, on any desk.

Around the example and the toolkit sits the practical machinery a working instrument needs: scenarios export to spreadsheets and to a session file that reloads with one click on a colleague's screen, and the online manual documents every convention this paper has touched.

These capabilities line up, deliberately, with the steps of an institutional projection round. The ECB's account of forecasting with its multi-country model (Angelini et al., 2025) decomposes the process into assessing the impact of external assumptions, updating the projection on new data, layering judgment through add-factors, and decomposing the revision for communication. The MFFM carries a disciplined analogue of each step: assumptions enter as conditioning paths, hard or soft; the projection updates through a data shift that re-anchors every forecast while the model stays fixed; judgment enters through the long-run anchors and the trust dial on outside forecasts, devices with stated mechanics rather than free add-factors; and a release-impact monitor records what each new quarter changed, bloc by bloc. An economist who knows the projection workflow will find its stations here, compressed from a committee cycle into a browser session.

Two lines of work come next. The out-of-sample forecast evaluation — the tool's model against the random walk, univariate autoregressions, a closed-economy Bayesian VAR, the classical GVAR, and a Bayesian GVAR in the manner of Crespo Cuaresma et al. (2016), in the tradition of Adolfson et al. (2007b) — has been restarted on the current vintage and will be reported in a companion paper. And the structural layer invites extension: generalized impulse responses and connectedness measures for the coupled system, and a broader menu of policy experiments from the model data base. The platform is built to absorb both without changing what a user already knows.

A Model detail

Section 2 carries the model itself; this appendix records the pieces a replicator needs beyond it.

The prior, exactly. Stacking bloc i's coefficients as $\beta_i = (c_i, \Gamma_i, B_{i1}, \ldots, B_{ip})'$, the conjugate prior is $\Sigma_i \sim \mathcal{IW}(\Psi_i,\, d_i)$ and $\mathrm{vec}(\beta_i)\,|\,\Sigma_i \sim \mathcal{N}\big(\mathrm{vec}(b_i),\, \Sigma_i \otimes \Omega_i\big)$, with b_i the random-walk mean (own first-lag one; zero on the fiscal-ratio columns), and Ω_i diagonal with entries $\lambda_i^2/(\ell^2 \psi_j)$ for the coefficient on lag ℓ of variable j, following Giannone et al. (2015). The scales ψ_j are fixed at univariate AR residual variances; the intercept and seasonal-dummy rows carry a diffuse variance. The sum-of-coefficients prior enters as the dummy observations of Doan et al. (1984) with tightness μ_i; on seasonal blocs the dummy observations carry zeros in the D_t columns. Hyperpriors: $\lambda_i \sim$ Gamma with mode 0.2 and standard deviation 0.4; $\mu_i \sim$ Gamma with mode 1 and standard deviation 1; both updated by random-walk Metropolis–Hastings on the completed-data marginal likelihood, proposals reflected into $[10^{-4}, 5]^2$. The estimator is `cgl_bvarV5.m` in the tool's repository, driven by `export_bvar_gvar24_loop.m`.

The operator M. An entry of M answers one question: if partner j's solved level paths deviate from baseline by one unit in one cell, by how much does bloc i's star path deviate? The star definitions (7)–(9) make the map from partner paths to star paths linear with weights (10); the bloc's conditional-forecast operator (the smoother of (11) with pinned stars) is affine; composing the two over all bloc pairs, horizon step by horizon step, fills M. Four extra row blocks carry the US-originated global prices, which the US bloc's own solution determines and every other bloc receives. The construction is `buildLinearSystemEndo` in `gvarSolve.ts`; the export-time inverse is `precompute_gvar_coupling.mjs`.

Algorithm fine print. Within the outer iteration the smoother's companion is held at the previous pass's posterior mean and common random numbers are reused across passes, so successive passes differ only through the star data — this is what makes the outer loop contract in practice. An exact data-augmentation pass (companion rebuilt at every Gibbs sweep) is available behind a flag as a final polish. Convergence is measured on the star histories excluding the pandemic-imputation window, whose cells legitimately churn across passes. The 17 euro-member satellites are estimated separately, each a standalone BVAR whose euro-area columns are pinned at solve time to the EA bloc's solution.

B Credibility: testing, interpretation, and a classical-GVAR comparison

A tool used for policy must show its work, and this appendix says how this one is tested, how its output should and should not be read, and where its evaluation stands.

Testing. The computation engine ships with an invariant test suite (about five hundred tests) that runs on every change: the TypeScript smoother is cross-validated against the MATLAB reference implementation on the canonical conditional forecast; the closed-form global solve is verified against the iterative fixed point; the seasonal forecast path produced by the filter and by the band simulator agree to numerical precision; and a deterministic-mean canary pins the shipped baselines to the estimation output. Data provenance travels with the data: every model-completed cell is flagged in the interface, vintages are archived on every data shift, and a release-impact monitor records what each new quarter changed. None of this substitutes for external review, which we invite.

Interpretation. A conditional forecast is a statement about likelihood, not about cause: it re-weights the model's futures by consistency with the pins (Waggoner and Zha, 1999), and only the structural layer of Section 3.4 makes causal claims, on the authority of the

DSGE model chosen there (Antolín-Díaz et al., 2021). The distinction is kept visible in the interface: the counterfactual is a separate line, labeled with its model.

The classical-GVAR confrontation. The one quantitative credibility exhibit we commit to in this paper is a same-data confrontation: a classical GVAR — estimated with the Smith–Galesi toolbox routines (Smith and Galesi, 2014) on the balanced window the classical recipe requires — subjected to the US credit-spread experiment, its generalized impulse responses (Pesaran and Shin, 1998) set against the tool's conditional deltas of Section 3. The exhibit is being produced with the restarted evaluation (below) and will appear in this appendix in the next revision.

A Bayesian-GVAR head-to-head. The nearest relative of our estimator is the Bayesian GVAR of Crespo Cuaresma et al. (2016), and we ran the comparison in both directions: their exercise replicated with their own archived code and data, and our estimator run on a common public panel. Their finding survives both. On their 36-country panel their code, at reduced posterior sampling over the same 2004–2013 hold-out, reproduces the structure of their results: shrinkage is what separates a useful GVAR from a useless one — at the year horizon the no-shrinkage GVAR's relative RMSE on short rates reaches 6.95 against 2.58 under the Sims–Zha prior — and the shrunk model beats the univariate AR(5) benchmark on output, equity, long rates and oil one quarter ahead. The working part is the prior family: a random-walk prior mean with sum-of-coefficients and dummy-initial-observation dummies, tuned by marginal likelihood — the same family this tool's estimator carries (Section 2). Run on the public 28-country `pesaranData` panel under the same expanding-window design, our G-BVAR delivers the same qualitative verdict against the AR(5) — and the full exercise, carried to nine quarters ahead across nine models with anchored and no-global-solve variants and a regional cut, is Appendix C, which mirrors their Section 4.3 paragraph by paragraph. The package implementation of the Bayesian GVAR estimated at its defaults — a white-noise prior mean, no sum-of-coefficients dummies — loses to the AR(5) on every variable on the same panel. That is a caution about defaults, not about the method: with the prior Crespo Cuaresma et al. actually use, their result stands, and our estimator inherits it.

Forecast evaluation. The out-of-sample evaluation has been restarted on the current vintage, in the simplest form that answers the question: the tool's model, with its seasonal treatment and with and without the shipped long-run anchors, against the random walk, an AR(1), a closed-economy Bayesian VAR with the same prior, the classical GVAR above, and the Bayesian GVAR of the head-to-head, across all coupled blocs — the benchmark tradition of Adolfson et al. (2007b). A second leg will follow the design the ECB uses for its own multi-country model (Angelini et al., 2025): condition on the *realized* financial and external paths and ask how well the model translates correct assumptions into macro forecasts, against a random-walk reference — an exercise our conditioning machinery performs natively — illustrated on one narrative episode. A four-origin pilot of the evaluation on the platform's own panel closes Appendix C; the full expanding-window results go to a companion paper — this one documents the platform.

C Forecast performance on the CCFH design

Two strands of literature meet in this exercise. The first studies Bayesian shrinkage as the route to forecasting with vector autoregressions: from the Minnesota prior of Doan et al. (1984) and Litterman (1986) and the weighted combination of its components in Sims and

Zha (1998), through the demonstration by Bańbura et al. (2010) that, with shrinkage tightened as the system grows, very large systems forecast as well as small ones, to Koop (2013) and Carriero et al. (2015) on specification and prior choices in medium and large systems, Giannone et al. (2015) on setting the shrinkage hyperparameters by marginal likelihood — the tuning approach our estimator carries — and Clark (2011) on the density side. The second asks whether modeling international linkages pays: the GVAR of Pesaran et al. (2004) and Dees et al. (2007) was first systematically evaluated out of sample by Pesaran et al. (2009), who report gains over univariate benchmarks that are present but uneven across variables and countries; Greenwood-Nimmo et al. (2012) extend the evaluation to probabilistic forecasts, Han and Ng (2011) to a regional panel, and Dovern et al. (2015) ask directly whether joint modeling of the world economy improves multivariate forecasts. Crespo Cuaresma et al. (2016) — henceforth CCFH — join the two strands: Bayesian shrinkage estimated within a GVAR, evaluated over a forty-quarter hold-out against univariate and non-shrinkage alternatives, with the conclusion that the combination — not either ingredient alone — delivers the improvement. Their design has become the reference for the class, and this appendix adopts it; Chudik and Pesaran (2016) survey the wider GVAR literature, and Adolfson et al. (2007b) established the corresponding benchmark-suite tradition for open-economy policy models.

The evaluation conventions are equally settled. Point accuracy is reported as root mean squared error relative to a naive univariate benchmark, density accuracy as log predictive scores — the measure grounded in Geweke (2001) and surveyed by Geweke and Whiteman (2006), with Geweke and Amisano (2010) on the comparison of predictive distributions — and differences in predictive accuracy are assessed in the tradition of Diebold and Mariano (1995) with the small-sample correction of Harvey et al. (1997). Three questions recur across this literature without resolution. Horizons beyond one year are rarely examined — most GVAR evaluations, CCFH included, stop at four quarters — although the shrinkage priors at issue are designed around low-frequency behavior. The respective contributions of the prior and of the international linkages are confounded in every comparison that changes both at once, as Pesaran et al. (2009) and Crespo Cuaresma et al. (2016) do by construction. Finally, the information in observable forward-looking data — policy guidance, the yield curve — enters none of the standard specifications, although the hold-outs of this literature span the effective-lower-bound years in which that information contradicted historical means most sharply. The exercise below addresses all three: it extends the horizon grid to nine quarters, holds the prior fixed while removing the linkages, and evaluates anchors built from each origin's own term structure.

The evaluation design matches CCFH's in all but four respects, following their Section 4.3 step by step. The initial estimation sample ends in 2003:Q4, and the 40 quarters 2004:Q1–2013:Q4 form the hold-out. Every model is re-estimated at each origin over an expanding window. Point forecasts are evaluated by root mean squared error and density forecasts by log predictive scores; all measures are benchmarked against a fifth-order univariate autoregression. Four differences remain. First, where they consider one- and four-quarter horizons, we evaluate every horizon up to nine and report $h = 1$, 5 and 9 ($h = 1$ denotes the nowcast quarter, so $h = 9$ corresponds to $t+9$ under origin-relative dating). Second, where their panel is the 36-country, seven-variable dataset of Dovern et al. (2015), ours is the public 28-country `pesaranData` panel with the six common variables (real output, inflation, short- and long-term interest rates, the real exchange rate, and equity prices; total credit is not available), and two lags rather than their five, which keeps the 40 re-estimations of 28 country models tractable. Third, we summarize by the cross-country median where their tables report unweighted averages. Fourth, the `BGVAR` package runs at its default prior hyperparameters — with one stated exception: the eigenvalue truncation is raised from 1.05 to 1.5, without

Table 2: Point-forecast accuracy on the CCFH design: relative RMSE against the AR(5) benchmark, median across the 28 `pesaranData` countries (40 origins, 2004–2013, re-estimated at every origin). A dagger marks cells whose summed log predictive score exceeds that of the AR(5). "+anch." applies the long-run targets of Section 4.2: drift targets for trending log-levels, the recursive as-of-origin mean for inflation, and, for the interest rates, the forward-rate path implied by each origin's own yield curve under the expectations hypothesis (see text). No future information enters any target. The SSVS and MN columns run the `BGVAR` package at default hyperparameters; RW is the last-observation random walk, point-scored only.

	G-BVAR	+anch.	no-solve	+anch.	cl. GVAR	SSVS	MN	RW
One quarter ahead (the nowcast, $h = 1$)								
Output	0.87†	0.89†	0.90†	0.91†	0.96†	1.85	1.90	1.10
Inflation	1.02†	1.07	1.01†	1.04	1.26	1.68	1.76	1.05
Short rate	0.95†	0.99†	0.99†	0.96†	1.60	6.24	10.50	0.97
Long rate	1.09	1.12	1.04	1.07	2.02	10.25	14.87	0.98
Exch. rate	0.99	0.97†	0.98†	0.96†	1.37	1.55	1.67	0.99
Equity	0.96†	1.05	0.95†	0.98†	2.37	1.57	1.67	1.01
Five quarters ahead ($h = 5$)								
Output	0.75†	0.74†	0.76†	0.73†	1.85	1.21	1.00	0.98
Inflation	1.00†	1.18	0.98†	1.12	1.70	2.38	1.59	1.13
Short rate	0.85†	0.87†	0.89†	0.90†	2.52	3.11	3.51	0.84
Long rate	0.96†	1.14	0.90†	1.11	4.75	4.34	5.40	0.80
Exch. rate	0.90†	0.78†	0.86†	0.78†	2.65	0.96	0.88	0.78
Equity	0.87†	0.94	0.85†	0.91†	2.88	1.32	1.09	0.90
Nine quarters ahead ($h = 9$)								
Output	0.64†	0.62†	0.62†	0.62†	1.50	1.15	0.87	0.93
Inflation	0.96†	1.12	0.95†	1.13	2.30	2.85	1.47	1.14
Short rate	0.72†	0.77†	0.80†	0.76†	1.83	2.41	2.37	0.76
Long rate	0.85†	1.06	0.78†	1.05	2.16	3.60	4.18	0.72
Exch. rate	0.79†	0.75†	0.81†	0.74†	2.45	0.93	0.77	0.73
Equity	0.84†	0.89	0.86†	0.90	4.23	1.36	1.01	0.86

which no SSVS draw survives on data in near-unit-root levels — whereas their specification tunes hyperparameters by marginal likelihood; their published configuration is reproduced with their archived code and data in Appendix B. One further departure is common to both sides: the trade weights are the package's 1980–2016 averages, where CCFH restrict weights to 1980–2003 to keep them real-time. The weights enter our leg and the package's leg identically, so the head-to-head is unaffected, but neither leg is strictly real-time in the weights. The two exercises therefore answer separate questions: this one compares the estimators under identical conditions; that one asks whether their published results can be reproduced.

Overall results, one quarter ahead. Their one-quarter findings are threefold: most specifications improve upon the naive benchmark in point forecasts; neither country-specific shrinkage nor international linkages dominates when considered separately; and combining the two improves forecast performance considerably. The first panel of Table 2 confirms the first finding for our estimator: the G-BVAR improves upon the benchmark for output (0.87), the short rate (0.95) and equity prices, is at approximate parity for inflation and the exchange rate (1.02 and 0.99), and underperforms only for the long rate. The classical GVAR attains

0.96 for output but exceeds unity for the remaining variables, and the package's ratios at default hyperparameters exceed unity everywhere — output 1.85 under SSVS, short rates 10.50 under the Minnesota prior — consistent with their observation that the conjugate prior without tuned shrinkage performs least well, here in more extreme form. On the second and third findings our design permits a sharper decomposition. Their comparison sets a country-specific BVAR against a GVAR under a diffuse prior, so the prior and the linkages change together; ours holds the prior fixed and removes only the foreign variables. The G-BVAR and no-global-solve columns of Table 2 differ by at most 0.08 at any variable and horizon: conditional on the prior, the international linkages contribute little to point accuracy on this panel. On this evidence, the improvement they find from combining shrinkage with linkages owes primarily to the shrinkage, which in their comparison arrives jointly with the linkages. The same division of labor is why the platform justifies the coupled solve by cross-country consistency rather than by forecast accuracy (Section 3).

Density forecasts. Their density results confirm their point results, with the largest improvements among the shrinkage GVARs. The same holds here: the sum over countries of the G-BVAR's log-score differences relative to the AR(5) is 5.1 for output at the nowcast and 7.8 at $h = 9$ (the corresponding cells carry daggers in Table 2); the no-global-solve column performs comparably; the package's densities are inferior throughout at default hyperparameters. For the interest rates the anchored columns match the unanchored densities once the rate targets are consistent with the yield curve — the design question taken up below — while their inflation and equity daggers are lost.

Longer horizons. At four quarters they find the naive benchmark difficult to beat, forecast gains less pronounced than at one quarter, and the Sims–Zha prior strongest in point accuracy — evidence that the advantage of the random-walk prior with sum-of-coefficients dummies grows with the horizon. Extending the grid confirms and strengthens this pattern: the G-BVAR's output ratio declines at every horizon on the grid, from 0.87 at the nowcast to 0.75 at $h = 5$ and 0.64 at $h = 9$, and the short-rate ratio reaches 0.72. The random walk is strong for interest rates at intermediate horizons — 0.84 against the AR(5) at $h = 5$ — but the G-BVAR matches it there and improves upon it at $h = 9$, while the best package column attains 0.87 for output. Their evidence ends at the horizon at which the prior family's advantage emerges; on the extended grid the G-BVAR's advantage increases through the final quarter.

Long-run anchors. This variant has no counterpart in their study and illustrates the importance of target selection. A first specification anchors every variable to its recursive as-of-origin sample mean, and the short-rate ratio rises to roughly twice the benchmark (2.03 at the nowcast, 2.10 at $h = 9$): a mean-reversion target estimated on three decades of declining rates implies a return to four or five percent, whereas forward guidance over the effective-lower-bound years indicated that rates would remain near zero.

Replacing the rate targets with the path priced by the market — the forward rates implied by each origin's own yield curve under the expectations hypothesis — removes almost all of the damage. That path is the information an overnight-indexed-swap (OIS) curve would provide, and the only real-time version of it in this panel. The anchored short rate attains 0.99 at the nowcast, 0.87 at $h = 5$ and 0.77 at $h = 9$ — marginally above the unanchored 0.95, 0.85 and 0.72 on the median — with a country mean below that of the unanchored model at the long horizon (0.75 against 0.85) and a higher log score (5.2 against 5.0). At an effective-lower-bound origin the curve is low and flat, so the anchor holds the forecast down, as the guidance implied. The long rate records the remaining cost — 1.06 against the unanchored 0.85 at $h = 9$ — the consequence of extrapolating a two-point curve without

Table 3: Cross-country differences, following the regional analysis of Crespo Cuaresma et al. (2016): median {interquartile range} of country-level relative RMSE against the AR(5), output, one quarter ahead. Latin America contains only Chile on this panel.

	G-BVAR	no-solve	cl. GVAR	SSVS	MN	RW
Europe (12)	0.84 {0.12}	0.88 {0.12}	0.94 {0.10}	1.76 {1.03}	1.67 {0.71}	1.01 {0.16}
Other dev. (5)	0.94 {0.09}	0.93 {0.04}	1.08 {0.15}	1.98 {0.41}	1.77 {0.53}	1.24 {0.15}
Em. Asia (8)	0.80 {0.04}	0.80 {0.07}	1.05 {0.37}	1.75 {0.73}	1.97 {0.83}	1.30 {0.59}
Lat. Am. (1)	1.00 {0.00}	1.08 {0.00}	1.26 {0.00}	1.86 {0.00}	2.21 {0.00}	1.36 {0.00}
MEA (2)	0.89 {0.06}	0.96 {0.02}	0.88 {0.05}	2.02 {0.55}	2.35 {0.10}	1.40 {0.40}

a term-premium model, and inflation, still anchored to the recursive mean, pays the same penalty as the mean-target rates in kind (1.18 against 1.00 at $h = 5$).

The conclusion is that anchors improve forecasts when the target embodies forward-looking information — official long-run targets in the live platform, yield-curve-implied paths for interest rates — and not when it is a mechanical sample moment. Anchoring the level of a trending variable to its historical mean fails severely (output ratios near six), and anchoring rates to their historical mean imposes the previous interest-rate regime on the forecast. The natural extension in the live platform is to take its rate anchors from observed OIS curves — a data requirement, not a modeling one.

Cross-country differences. Their regional analysis finds that point-forecast accuracy varies more across regions than across variables: Latin America shows the widest dispersion and the least accurate forecasts; European and other developed economies are homogeneous, with tight cross-sectional distributions; and the distributions tighten when the priors incorporate country-specific shrinkage. Table 3 confirms the tightening on our side: the G-BVAR's output medians are 0.84 in Europe and 0.80 in emerging Asia, with interquartile ranges of 0.12 and 0.04, while the package's dispersion is between four and twenty times wider (1.03 against 0.12 for SSVS in Europe, 0.73 against 0.04 in emerging Asia). Their Latin American finding cannot be re-examined on this panel, which contains only Chile (1.00, at parity with the benchmark); it is covered instead by the reproduction with their own data in Appendix B, an exercise whose hold-out spans the same 2004–2013 crisis window.

The platform's own panel. The same protocol, transplanted to the panel the platform ships — 38 blocs, the V3 estimation configuration, the random walk as the naive benchmark — closes the appendix; Table 4 reports the four-origin pilot the evaluation protocol requires before its full expanding-window run, which goes to the companion paper. Three observations carry. First, inflation is where the platform's model earns its keep against the naive benchmarks at short and medium horizons (0.84 at the nowcast, 0.81 at $h = 5$), and its density forecasts improve on the random walk almost everywhere at the nowcast. Second, the no-coupling country BVAR matches or betters the coupled model in nearly every cell — the pesaranData ablation's conclusion, reproduced on the platform's own panel: the coupling purchases cross-country consistency, not point accuracy. Third, at this pilot's scale the univariate benchmarks remain the strongest long-horizon point forecasters — the AR(5) reaches 0.52 for inflation and 0.85 for growth at $h = 9$ — an expected outcome for four origins that straddle the pandemic, and the question the full expanding window is designed to settle rather than this gate. The shipped anchors are close to neutral for growth and inflation; on the policy rate they cost the coupled model at the nowcast (1.30 against 0.97) and help at the long horizon (1.25 against 1.33), with a small benefit to the country model's rates

Table 4: The platform's own panel: relative RMSE against the random walk, aggregated over the six headline economies (US, euro area, Japan, UK, Canada, China) and the pilot's four origins (2016, 2019, 2022 and 2024, all Q1), shipped V3 configuration (seasonal treatment, COVID-as-missing). A dagger marks cells whose CRPS also beats the random walk's. "+anch." applies the shipped long-run anchors. The classical-GVAR column is diagnostic only: its unrestricted VECM forecasts are explosive on this near-unit-root panel (the stationarity guard logs, but does not repair, the affected windows). At $h = 8$ and 9 the 2024 origin's outcomes are not yet observed, so those aggregates cover three origins.

	G-BVAR	+anch.	ctry BVAR	+anch.	AR(5)	AR(1)	RW-drift	cl. GVAR
One quarter ahead (the nowcast, $h = 1$)								
GDP growth	$1.34^\dagger$	$1.39^\dagger$	$1.32^\dagger$	$1.32^\dagger$	$1.33^\dagger$	$1.10^\dagger$	$0.93^\dagger$	$6 \cdot 10^2$
Inflation	$0.84^\dagger$	$0.85^\dagger$	$0.86^\dagger$	$0.86^\dagger$	$0.72^\dagger$	$1.02^\dagger$	$1.04^\dagger$	$4 \cdot 10^2$
Policy rate	$0.97^\dagger$	1.30	$0.74^\dagger$	$0.74^\dagger$	$0.87^\dagger$	1.03	5.69	$9 \cdot 10^2$
Five quarters ahead ($h = 5$)								
GDP growth	$0.99^\dagger$	$0.99^\dagger$	$0.99^\dagger$	$0.99^\dagger$	$0.98^\dagger$	$0.99^\dagger$	$0.98^\dagger$	75.13
Inflation	$0.81^\dagger$	$0.80^\dagger$	$0.77^\dagger$	$0.77^\dagger$	$0.65^\dagger$	$0.78^\dagger$	$0.77^\dagger$	$4 \cdot 10^2$
Policy rate	1.11	1.07	$0.87^\dagger$	$0.85^\dagger$	$0.86^\dagger$	0.98	0.92	$5 \cdot 10^2$
Nine quarters ahead ($h = 9$)								
GDP growth	1.00	1.09	$0.98^\dagger$	$0.97^\dagger$	$0.85^\dagger$	$0.87^\dagger$	$0.87^\dagger$	$4 \cdot 10^2$
Inflation	$1.08^\dagger$	$1.08^\dagger$	$0.73^\dagger$	$0.69^\dagger$	$0.52^\dagger$	$0.64^\dagger$	$0.66^\dagger$	$5 \cdot 10^2$
Policy rate	1.33	1.25	1.04	$0.99^\dagger$	$0.85^\dagger$	1.00	$0.88^\dagger$	$1 \cdot 10^3$

at $h = 9$ (0.99 against 1.04). Two accounting notes: the four blocs whose post-re-sourcing series begin too late for the early origins (Australia, Denmark, Indonesia, New Zealand) are excluded from the coupled model's estimation at those origins by a data-availability rule — none is a headline economy, so the aggregates in Table 4 are unaffected — and the pilot is reported for completeness, not as evidence of forecast superiority: the platform's case rests on the conditional-scenario machinery of Section 3, with the CCFH-design results above showing the estimator competitive where designs are comparable.

Summary. Their three principal conclusions carry over, the first two with a refinement. First, where they conclude that international linkages and locally induced shrinkage jointly improve forecasts, our controlled comparison attributes the accuracy to the prior family and assigns the linkages their distinct role: internal consistency of the cross-country scenario, the property on which the platform rests. Second, where they find that no single shrinkage prior dominates, the package-at-defaults columns show the complementary result: at the one-quarter horizon every prior in the package loses to the univariate benchmark on every variable, so the marginal-likelihood tuning they emphasize is necessary for the package to compete. Our estimator carries the same prior family with the same tuning approach. Third, their regional heterogeneity is confirmed, including the tightening of cross-sectional dispersion under country-specific shrinkage. On their design and their metrics, the estimator behind this platform delivers the most accurate point forecasts for output, equity prices and, at most horizons, the short rate and the exchange rate; the long rate remains with the random walk (0.72 at $h = 9$ against the family's best 0.85). Its output advantage widens at every horizon on the grid, to 0.64 at $t + 9$ — a result their four-quarter horizon could not reveal.

D Coverage and reproducibility

Table 5 lists the coupled blocs with their sizes, samples, and blocks; the per-variable rosters, sources, and transformations for all 55 economies are in the online manual, and for the United States in Crump et al. (2025).

Table 5: The coupled blocs: size, sample, and blocks.

Bloc	n	Estimation sample	Seasonal dummies	Fiscal block
US	40	1998Q3–2025Q4	–	yes
EA	25	1999Q1–2025Q4	yes	yes
JP	25	2000Q1–2025Q4	yes	yes
UK	27	1997Q1–2025Q4	yes	yes
CA	27	1998Q1–2025Q4	–	yes
CN	22	1999Q1–2025Q4	yes	yes
CH	20	2000Q1–2025Q4	–	–
KR	25	2000Q1–2025Q4	yes	yes
MX	26	2001Q4–2025Q4	yes	yes
AU	25	1999Q1–2025Q4	–	yes
TR	23	2000Q1–2025Q4	yes	yes
SE	26	2000Q1–2025Q4	–	yes
PL	25	2000Q1–2025Q4	–	yes
NO	24	2000Q1–2025Q4	yes	yes
DK	26	2000Q1–2025Q4	–	yes
IN	21	2011Q4–2025Q4	yes	–
BR	24	2012Q2–2025Q4	–	yes
RU	17	2003Q1–2025Q4	–	–
ID	24	2005Q3–2025Q4	yes	yes
SA	18	2010Q1–2025Q4	–	–
AR	20	2004Q1–2025Q4	–	–
ZA	25	2002Q1–2025Q4	yes	yes
TH	26	2003Q1–2025Q4	–	yes
MY	21	2015Q1–2025Q4	–	–
CL	25	2000Q1–2025Q4	yes	yes
CO	26	2005Q1–2025Q4	yes	yes
CR	20	2010Q3–2025Q4	yes	yes
CZ	25	2000Q1–2025Q4	–	yes
HU	25	2000Q1–2025Q4	–	yes
IL	25	2000Q1–2024Q4	yes	yes
IS	21	2000Q2–2025Q4	yes	–
NZ	25	2000Q1–2025Q4	–	yes
TW	20	1997Q1–2025Q4	–	–
HK	17	2000Q1–2025Q4	yes	–
SG	15	2000Q1–2025Q4	–	–
RO	18	2000Q1–2025Q4	–	–
PE	20	2000Q1–2025Q4	yes	–
PH	17	2000Q1–2025Q4	–	–

Reproducibility. Every response figure and the numbers it quotes are generated by scripts in the tool's repository and imported here as macros; the coverage and dating constants are read from the shipped artifacts. From the repository root:

```
cd app
npx vite-node scripts/paper_figs/us_spread_cf.mjs    # Figs 1,3
npx vite-node scripts/paper_figs/global_spread.mjs  # Fig 2
```

```
npx vite-node scripts/paper_figs/nfa_pilot.mjs      # Fig 4
node scripts/paper_figs/rho_margin.mjs              # rho consts
node scripts/paper_figs/ccfh_headtohead.mjs         # App-B macros
node scripts/paper_figs/ccfh_matrix.mjs             # App CCFH tables
cd ../docs/paper
pdflatex mffm_platform_paper && bibtex mffm_platform_paper
```

The CCFH-design appendix additionally consumes, from the evaluation repository's `ccfh_replication/` directory, `region_score.R` (the regional cut) and `re_anchor_v2.py` (the drift-target anchored variants), both run against the merged forecast store documented there.

The Bayesian-GVAR comparison itself lives in the evaluation repository, in its `ccfh_replication/` directory: the BGVAR-package and G-BVAR expanding-window runs with their scoring scripts, and the archived Crespo Cuaresma et al. code and data with the driver that reproduces their hold-out; the `RESULTS` files parsed above are their durable record.

The interface views are screen captures of the deployed tool. The coverage table above is generated from the shipped data manifests by the script recorded in its header comment.

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